LAHORE UNIVERSITY OF MANAGEMENT SCIENCES
Department of Electrical Engineering

AI 501 Mathematics for Artificial Intelligence
Quiz 3 Solutions

Name: _____

Campus ID: _____

Total Marks: 10
Time Duration: 20 minutes

---

1. [2 marks] **Specificity and Recall – Definitions:**

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Positives}}, \qquad \text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}.$$

For each of the following cases, state which metric (**specificity** or **recall**) should be prioritized, and briefly justify.

(a) Cancer detection test.

(b) Spam email filter.

Solution:

- Cancer detection — prioritize **recall**, since missing a positive case is dangerous.
- Spam filter — prioritize **specificity**, to avoid flagging genuine emails as spam.

2. [2 marks] **Ranks and Spaces**

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

(a) Determine the **rank** of A. Briefly justify your answer.

(b) Based on the rank, state whether the columns of A are linearly independent or dependent, and explain what this means for the column space.

Solution:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}.$$

The second row is twice the first, so rows 1 and 2 are linearly dependent. Row 3 is not a multiple of row 1, so there are two independent rows.

$$\text{Rank}(A) = 2.$$

Since the rank is less than the number of columns (3), the columns are **linearly dependent**. This means the column space is 2-dimensional — all columns lie in the same plane in $\mathbb{R}^3$.

3. [6 marks] **Interpreting a Linear Regression Fit**

You fit a simple linear regression model $y = \beta_0 + \beta_1 x$ using the following dataset:

x	y
1	2
2	3
3	5

(a) Formulate this regression problem in the matrix form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$ by explicitly writing the design matrix $\mathbf{X}$, the parameter vector $\boldsymbol{\beta}$ and the response vector $\mathbf{y}$.

Solution: The linear model $y = \beta_0 + \beta_1 x$ for each data point i is:

$$y_1 = \beta_0 + \beta_1(1)$$
$$y_2 = \beta_0 + \beta_1(2)$$
$$y_3 = \beta_0 + \beta_1(3)$$

In matrix form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$, we have:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

(b) Using the least-squares formulation, compute the least-squares estimates $\hat{\boldsymbol{\beta}} = \mathbf{X}^{\dagger}\mathbf{y}$

Solution: The least-squares solution is given by $\hat{\boldsymbol{\beta}} = \mathbf{X}^{\dagger}\mathbf{y}$, where $\mathbf{X}^{\dagger}$ is the Moore-Penrose pseudoinverse.

Step 1: Compute $\mathbf{X}^T\mathbf{X}$

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

The determinant is:

$$\det(\mathbf{X}^T\mathbf{X}) = (3)(14) - (6)(6) = 42 - 36 = 6$$

Thus,

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{6}\begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix}$$

For a full-column-rank matrix, $\mathbf{X}^+ = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$:

$$\mathbf{X}^+ = \begin{bmatrix} \frac{7}{3} & -1 \\ -1 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$\mathbf{X}^+ = \begin{bmatrix} \frac{4}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

$$\hat{\boldsymbol{\beta}} = \begin{bmatrix} \frac{4}{3} & \frac{1}{3} & -\frac{2}{3} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

$$\boxed{\hat{\beta}_0 = \frac{1}{3}, \quad \hat{\beta}_1 = \frac{3}{2}}$$

(Intuitively, the line can also be estimated graphically — the data trend suggests a slope of about 1.5 and an intercept near 0.3.)

(c) Write down the fitted regression line and use it to predict y when $x = 4$.

Solution:

The fitted regression line is: $\hat{y} = \frac{1}{3} + \frac{3}{2}x$

Prediction for $x = 4$: $\hat{y} = 0.33 + 1.5(4) = 6.33$.